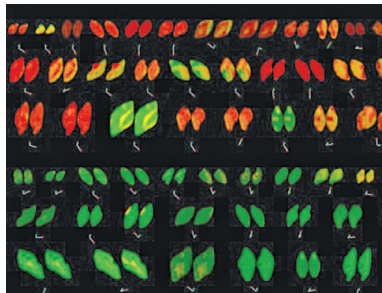


Pixel-level analysis reveals lithium-ion battery insights

Researchers at MIT, Stanford University, SLAC National Accelerator, and the Toyota Research Institute have made



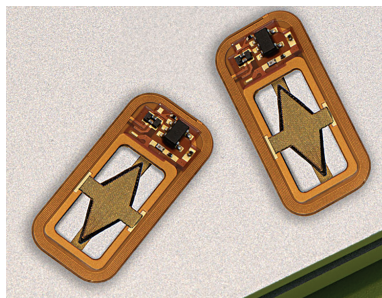
By analysing X-ray images of lithium iron phosphate particles as they charged and discharged, researchers have shown that the movement of lithium ions within the material was nearly identical to computer simulations they had created earlier. In each pair, the actual particles are on the left and the simulations are on the right (Courtesy of the researchers)

significant discoveries about the reactivity of lithium iron phosphate, a crucial material used in electric car batteries and other rechargeable energy storage systems, by analysing X-ray images. Leveraging 180,000 pixels as data points, the researchers developed a computational model to formulate precise equations describ-

ing the battery material's non-equilibrium thermodynamics and reaction kinetics. Additionally, they observed that the patterns of lithium-ion flow indicated potential spatial variations in lithium-ion absorption rates on different parts of the particle's surface. The study's implications suggest that optimising the thickness of the carbon layer on the electrode surface could lead to improved battery efficiency. Beyond its relevance to battery materials, the research team anticipates applying their analysis to investigate pattern formation in various chemical and biological systems.

A device promises diabetes management without needles

Engineers at MIT have developed a new implantable gadget that houses hundreds of thousands of islet cells capable of producing insulin, coupled with an inbuilt oxygen



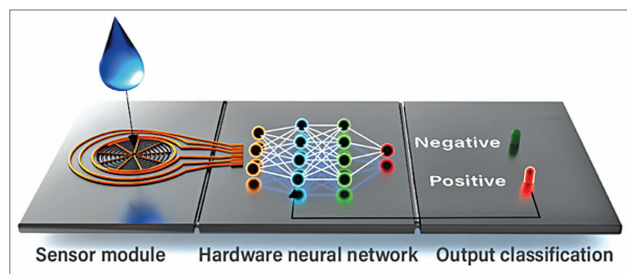
MIT engineers designed an implantable device that carries hundreds of thousands of islet cells along with its own on-board oxygen factory to keep the cells healthy (Credit: Felice Frankel)

generation system. The team has pioneered a method to indefinitely generate oxygen via water splitting, leveraging a proton-exchange membrane originally designed for hydrogen generation in fuel cells. Further advancements include a device,

roughly the size of a US quarter, proven effective in aiding diabetic mice. When tested, the device with the oxygen-generating membrane maintained regular glucose levels, unlike its non-oxygenated counterpart. Despite fibrosis challenges around implants, this device effectively regulated blood glucose levels, suggesting efficient insulin and glucose exchange. The method also shows promise for delivering cells to produce other therapeutic proteins, evidenced by its preservation of cells creating erythropoietin, vital for red blood cell stimulation.

Self-training neuromorphic biosensors

Researchers from Eindhoven University of Technology and Northwestern University in the US have developed a new neuromorphic biosensor with integrated learning capabilities, eliminating external training. The team used cystic fibrosis tests to assess their breakthrough chip. Employing



The neuromorphic biosensing chip (Credit: Eveline van Doremale)

the standard sweat test method, where elevated chloride anion levels suggest the disorder, they utilised their on-chip learning sensor. The chip's neuromorphic biosensor, consisting of a sensor unit, neural network hardware, and an output classifier, identified ion concentrations via ion-specific electrodes. Results were displayed as green or red lights, denoting negative or positive outcomes. Importantly, before its primary use, the neural network underwent supervised 'data gym' training, marking the innovation of the chip's self-training capability without external software. Additionally, the chip's design permits retraining for varied applications, introducing the possibility of personal user-trained implantable neural networks using direct individual data.

A scalable method for crafting ultra-thin protonic ceramic cells

Researchers from Kansas State University, Colorado School of Mines, Nissan Technical Center North America (NTCNA), Nissan Motor Company, and the University of Oklahoma have developed a composite positive electrode to significantly reduce various resistances in protonic